Retail forecasting research implementation supplement

This supplement extends the preserved hourly study with direct upcoming-day forecasting, richer calendar proxies, department behavioural analysis, a hybrid ensemble, stability checks and operational scenarios. Read it alongside research_report.md for the original data audit and one-hour results. Daily and hourly scores use different targets and must not be ranked together.

The common daily test contains 105 recorded shop days. The lowest daily RMSE is Prophet_With_Payday: GBP 237.650, with MAE GBP 184.163. Rankings are holdout-specific. The observed target is recorded revenue, not independently verified complete demand.

Forecast task and data integrity

Each forecast is issued at 00:00 Europe/London on its target day, before any target-day transactions are observed, and covers that local day. Past completed days may enter lag histories at later origins. The daily horizon is 23, 24 or 25 elapsed hours around daylight-saving changes. LSTM steps represent local calendar days. Daily sales sum existing net-sales observations; missing hours are not filled with zero. Unknown full days remain missing and cannot become target zeros. No assertion of complete shop-day capture is possible without opening-hours and till-system evidence.

The training period is 2025, validation January to March 2026, and test April 1 to July 14 2026. The first 28 days are excluded consistently to establish lag context. Imputers and target scalers fit training only. Frozen base models never train on test. Original CSVs and hourly outputs remain intact. Derived department aggregates exclude direct customer/receipt/payment identifiers; this does not certify raw-data anonymisation.

Calendar and payday methods

England was confirmed by the researcher. Bank holidays use the cached official GOV.UK England-and-Wales JSON snapshot, including substitute dates. Working-day means Monday-Friday excluding these holidays, not verified store opening. Both paired variants include holidays and working-day features. Payday variants alone add three flags: first/last three calendar days of a month; Friday and Saturday as an illustrative weekly salary-spending window; last working day of each month plus two following days. Last-working-day moves backwards over weekends and bank holidays. This is a configurable payroll scenario, not an established public-sector payroll rule. Actual customer payday and employer schedules remain unavailable. Overlapping proxy flags are allowed.

Model settings and hybrid design

Prophet uses daily targets with weekly seasonality, monthly cyclic regressors, lag/rolling predictors and holiday indicators; yearly seasonality is disabled. A shared paired validation choice selects changepoint prior scale from 0.01 and 0.05. XGBoost uses 200 trees, learning rate 0.03, minimum child weight 5, two CPU threads, seed 42, and a shared depth choice from 2 and 3. Past recorded-hours coverage is a lagged feature, never target-day coverage.

LSTM uses seven preceding calendar days of scaled sales, observed masks and calendar features, plus known target-day calendar inputs. One 16-unit LSTM feeds a 16-unit ReLU dense layer and linear output. Adam learning rate 0.001, batch size 32, maximum 40 epochs, patience 5, chronological unshuffled training, restored best validation weights. A training-only target scaler is inverted for GBP predictions. Seeds 42, 43 and 44 are all reported; seed 42 is the preselected primary model, not chosen using test accuracy.

Hybrid is the preserved convex ensemble of Prophet, XGBoost and LSTM. Shared nonnegative weights summing to one minimise average paired validation squared loss. Validation forecasts are out-of-training-sample for base models but in-sample for weight fitting. Sequential_Hybrid is now implemented separately as Prophet plus an XGBoost residual correction plus an LSTM correction of remaining residuals. Its disjoint January-June, July-September and October-December training blocks prevent later stages learning on earlier-stage in-sample residual targets. See research_completion.md for settings and stage diagnostics. The cascade is a later exploratory addition after earlier test results were known; it is not a new blinded confirmatory experiment.

model	weight
Prophet	0.273
XGBoost	0.088
LSTM	0.638

Verified common daily test results

model	n	MAE	RMSE	MAPE positive only
Prophet_Without_Payday	105	185.174	238.652	19.738
Prophet_With_Payday	105	184.163	237.650	19.812
XGBoost_Without_Payday	105	324.707	435.542	25.799
XGBoost_With_Payday	105	325.701	437.262	25.647
LSTM_Without_Payday	105	264.470	350.557	22.607
LSTM_With_Payday	105	263.162	350.361	22.380
Hybrid_Without_Payday	105	237.738	311.218	21.108
Hybrid_With_Payday	105	234.935	309.118	20.799
Sequential_Hybrid_Without_Payday	105	280.711	344.967	31.876
Sequential_Hybrid_With_Payday	105	265.494	328.060	29.907
Naive	105	266.913	333.495	26.813
Weekly_Seasonal_Naive	105	199.142	253.146	19.424

MAE/RMSE retain full numerical precision in CSVs; only report display is rounded. MAPE is reported only on strictly positive recorded daily targets with its denominator count. It is not a general net-sales accuracy measure and can exaggerate low-sales-day errors. Baselines use previous-day or previous-week recorded revenue; unavailable lag values fall back to the training median.

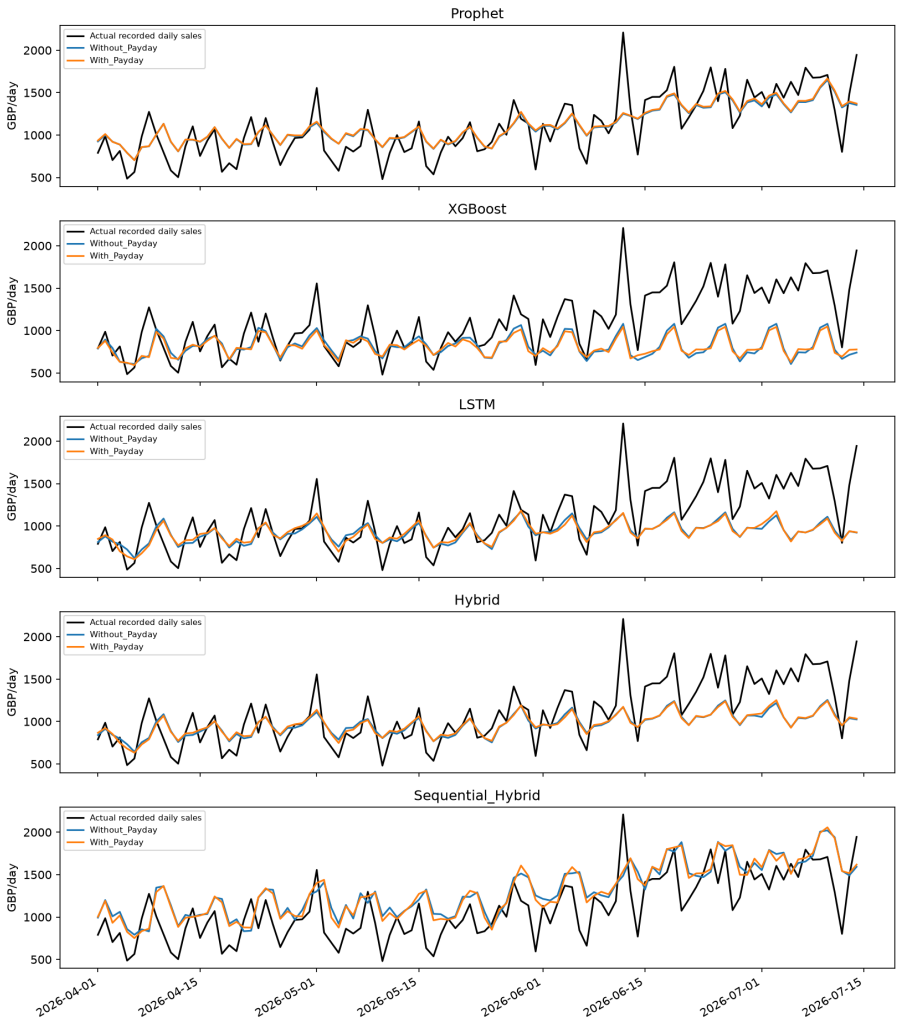


Figure 1. Upcoming-day actual recorded net revenue and paired forecasts, GBP per local day.

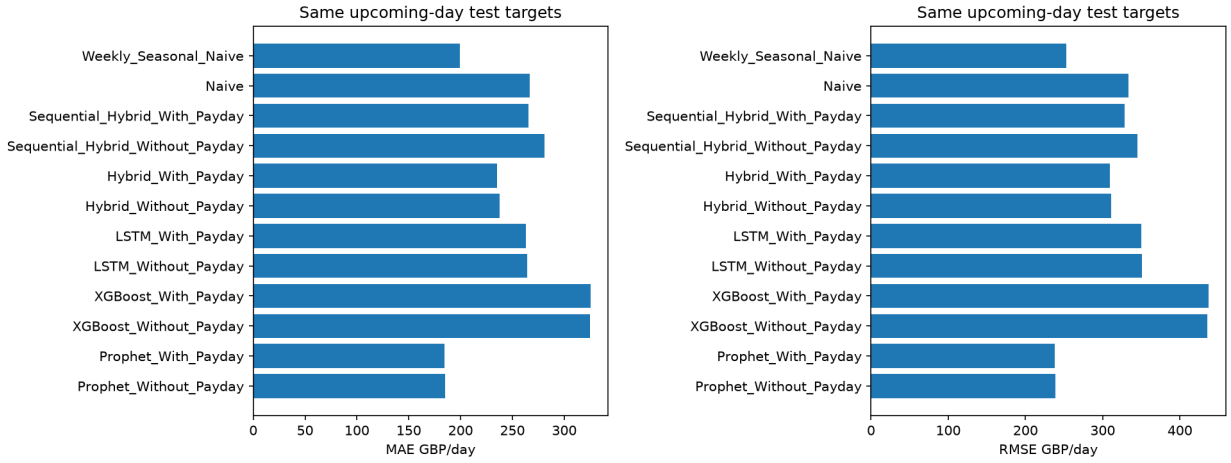


Figure 2. MAE and RMSE on identical recorded daily targets; includes baselines and hybrid.

Payday effects and subgroups

family	group	n	MAE change	RMSE change
Prophet	all	105	-1.011	-1.003
Prophet	payday	21	0.844	0.724
Prophet	non_payday	84	-1.474	-1.333
XGBoost	all	105	0.994	1.720
XGBoost	payday	21	0.649	0.082
XGBoost	non_payday	84	1.080	2.088
LSTM	all	105	-1.308	-0.196
LSTM	payday	21	-8.790	-11.753
LSTM	non_payday	84	0.563	2.119
Hybrid	all	105	-2.803	-2.100
Hybrid	payday	21	-9.637	-9.738
Hybrid	non_payday	84	-1.095	-0.613
Sequential_Hybrid	all	105	-15.216	-16.907
Sequential_Hybrid	payday	21	-21.701	-14.793
Sequential_Hybrid	non_payday	84	-13.595	-17.396

Change is error with payday minus error without payday; negative means improvement. Payday/non-payday subgroup labels use the original monthly flag so the partition stays consistent across all variants. Weekly and working-day proxies may overlap either group; their individual contribution is explained by SHAP, not isolated causal effects.

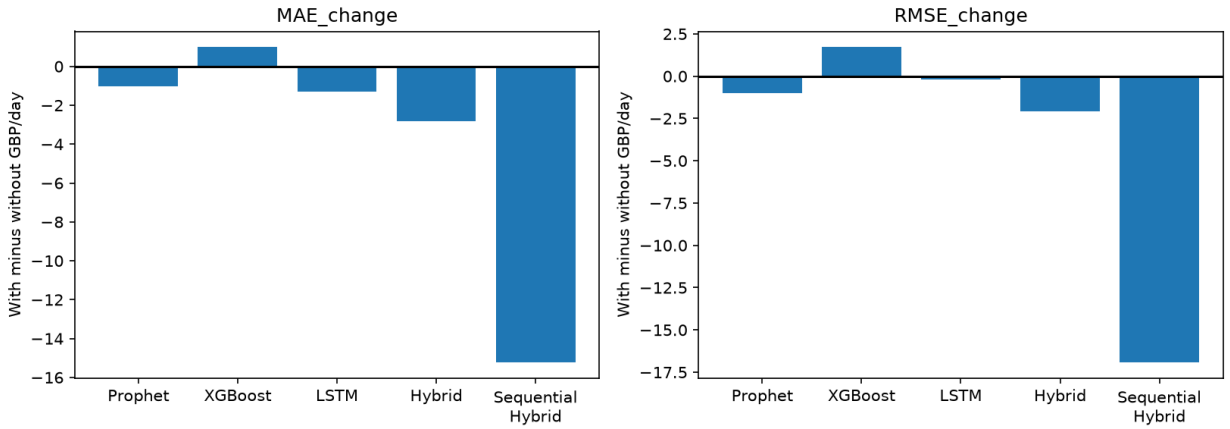


Figure 3. Adding the three calendar payroll proxies can improve or worsen errors; results determine the direction.

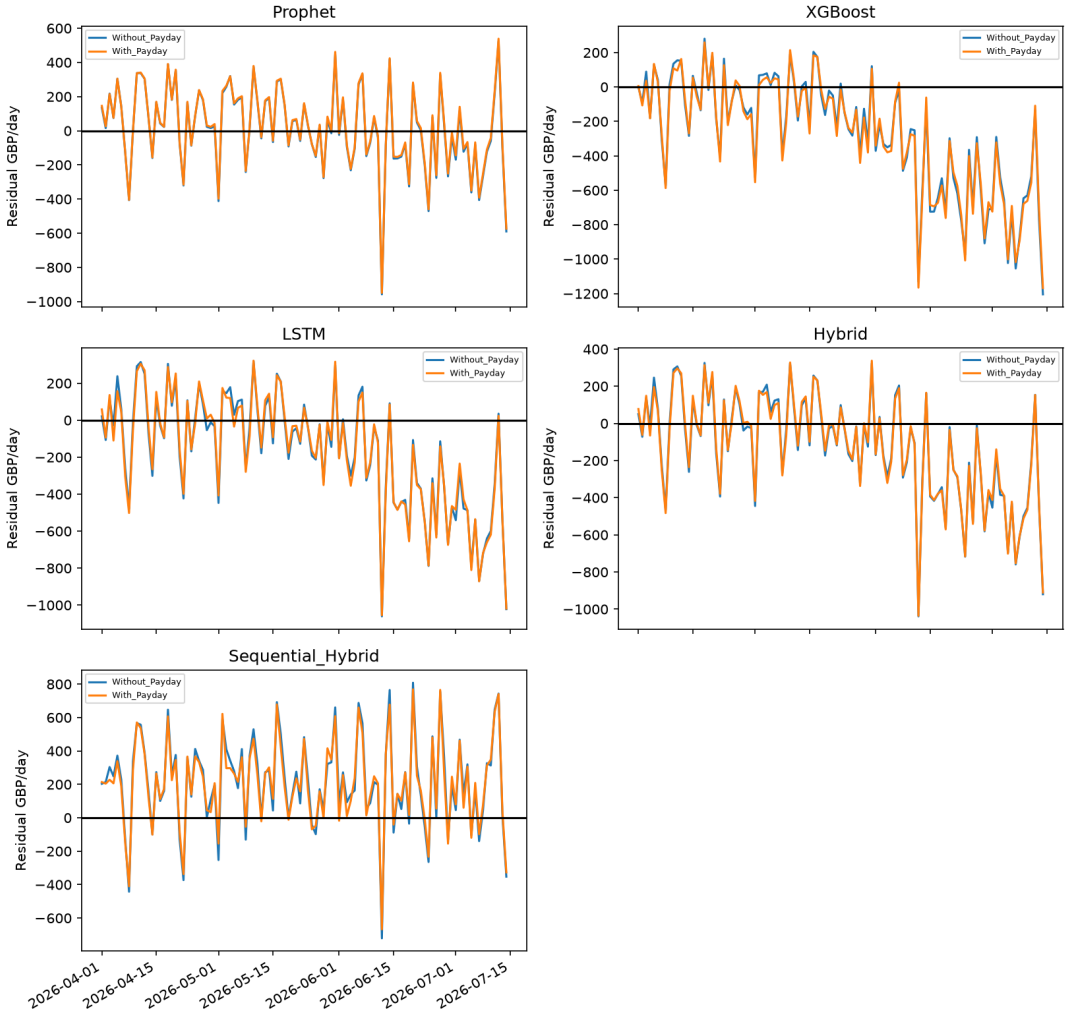


Figure 4. Daily signed residuals show temporal underprediction and overprediction.

Temporal uncertainty and stability

family	block days	mean with minus without	ci 2 5	ci 97 5
Prophet	7	-1.011	-3.024	1.008
Prophet	14	-1.011	-3.470	1.300
XGBoost	7	0.994	-3.461	5.422
XGBoost	14	0.994	-4.097	5.273
LSTM	7	-1.308	-6.476	3.589
LSTM	14	-1.308	-6.134	3.542
Hybrid	7	-2.803	-6.878	0.758
Hybrid	14	-2.803	-6.295	0.877
Sequential_Hybrid	7	-15.216	-21.414	-9.656
Sequential_Hybrid	14	-15.216	-21.899	-9.106

Paired daily loss differences use 1,000 circular moving-block bootstrap resamples with 7-day and 14-day blocks. Intervals concern mean absolute or squared loss differences, not RMSE differences. The approach preserves dependence within blocks but assumes approximately stable loss behaviour and omits dependence between blocks. Drift, few monthly cycles

and model selection uncertainty limit interpretation; intervals are descriptive sensitivity evidence, not strong causal or universal significance claims. Test days must be contiguous; the comparison log checks this.

payday	validation start	n	MAE	RMSE
False	2025-10-01	92	119.306	144.598
True	2025-10-01	92	116.910	141.947
False	2026-01-01	31	156.029	207.889
True	2026-01-01	31	151.692	201.709
False	2026-02-01	28	125.886	146.522
True	2026-02-01	28	129.778	151.820
False	2026-03-01	31	205.354	254.400
True	2026-03-01	31	202.797	252.681

Expanding-window validation evaluates fixed XGBoost settings before the April test boundary. Monthly test errors are saved for every family, and all three LSTM seeds are reported. These checks cover more than deterministic reruns but do not constitute nested cross-validation of every family. Payday-width sensitivity uses validation only and does not retroactively alter the reported test experiment.

model	n	MAE	RMSE
LSTM_Without_Payday	105	264.470	350.557
LSTM_With_Payday	105	263.162	350.361
LSTM_Without_Payday_seed_43	105	258.121	339.799
LSTM_With_Payday_seed_43	105	254.175	336.500
LSTM_Without_Payday_seed_44	105	251.202	327.862
LSTM_With_Payday_seed_44	105	258.966	344.585

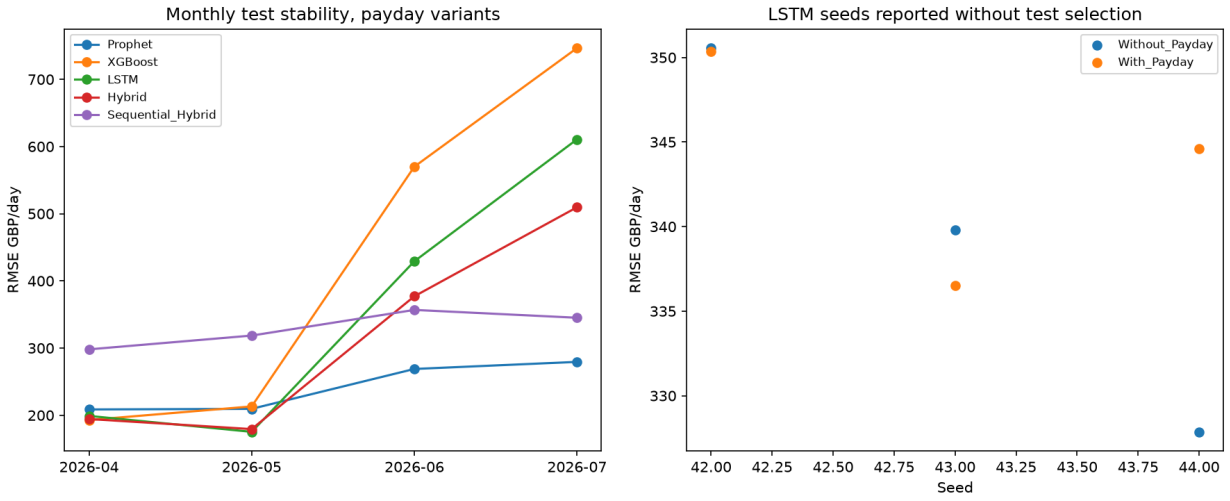


Figure 5. Monthly daily test errors and all predeclared LSTM seeds show performance variability; no test seed is selected.

month boundary width	n	MAE	RMSE
1	90	153.924	198.593
3	90	154.935	199.511
5	90	154.935	199.511

Department patterns and explainability

Hourly and weekday profiles use signed transaction totals, retaining receipt adjustments in an explicit unknown/adjustment group rather than reallocating them to products. Department-day tables measure recorded sales on recorded shop days; no category transactions imply no recorded category sales, not proof of zero demand. Tables contrast monthly, weekly and last-working-day indicators by category. Unadjusted group differences mix weekday composition, trend, promotions and recording coverage and cannot establish payday causation.

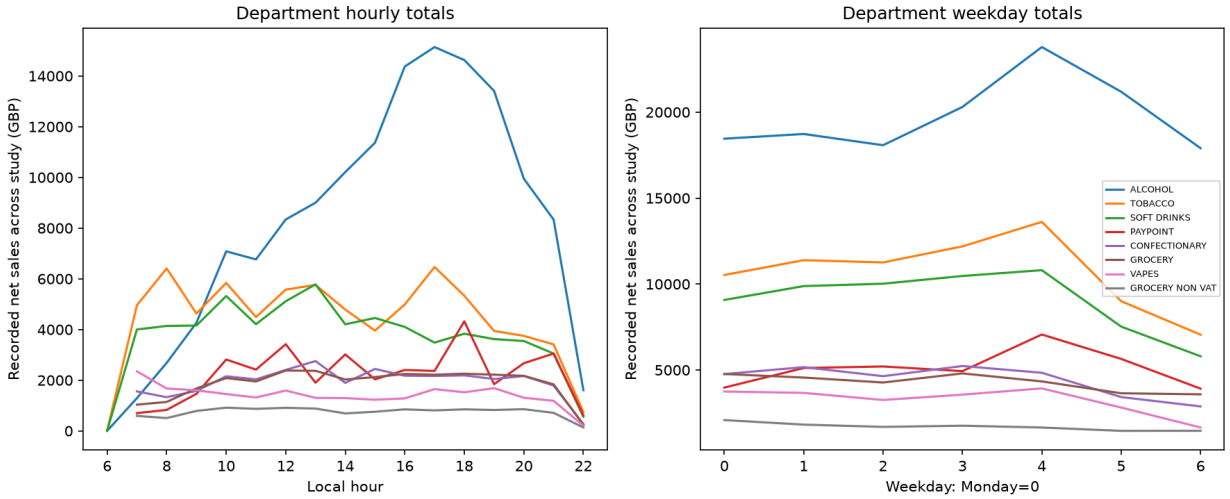


Figure 6. Largest recorded-revenue categories by local hour and weekday; totals also reflect observation coverage.

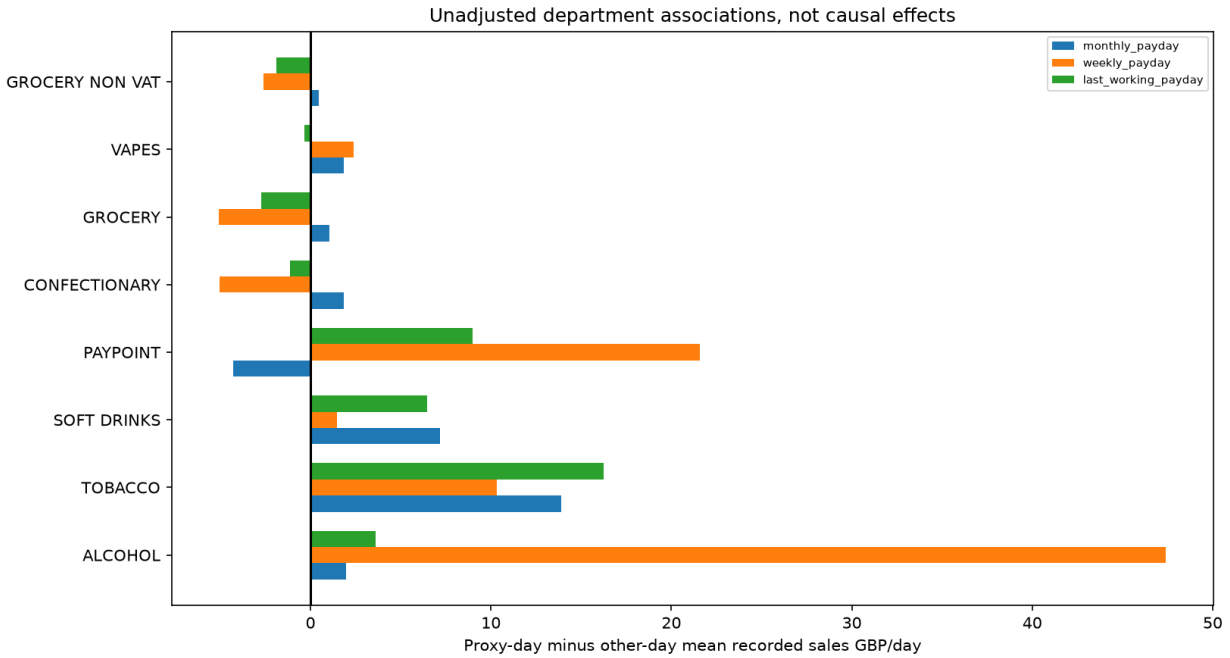


Figure 7. Department payroll-proxy group differences are descriptive associations, not causal customer-payday effects.

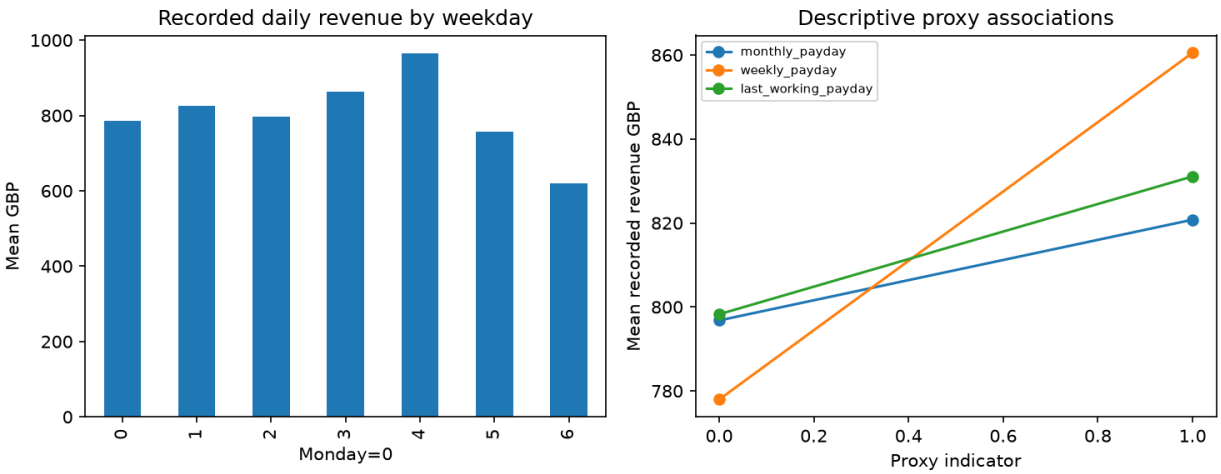


Figure 8. Daily weekday means and descriptive payroll-proxy associations.

feature	mean absolute shap
rolling_7	50.971
dow_cos	44.924
working_day	25.924
lag_1	24.773
lag_7	24.164
lag_28	19.177
trend_days	13.353
lag_14	12.572

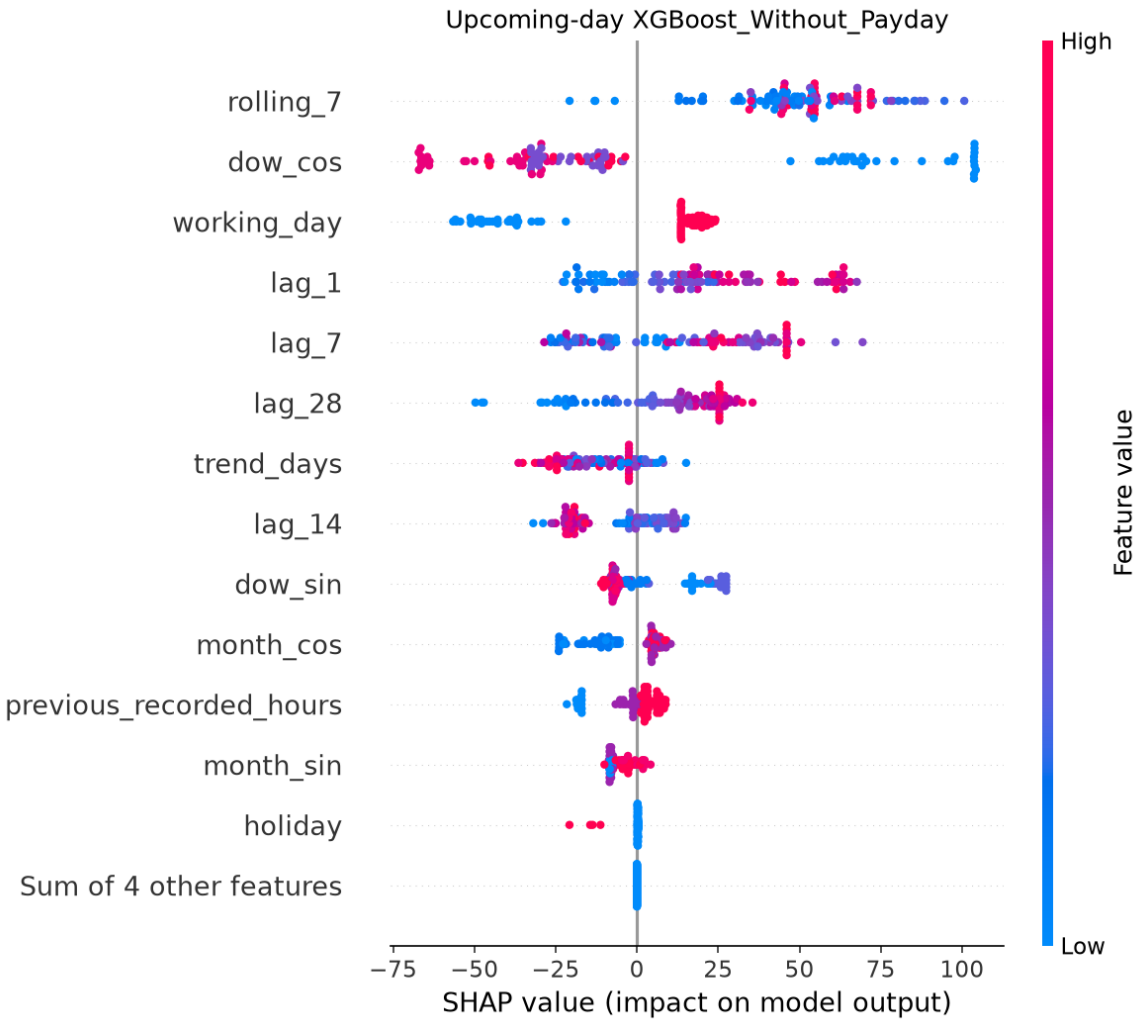


Figure 9. Daily XGBoost Without_Payday Tree SHAP contribution distributions in GBP.

feature	mean absolute shap
rolling_7	54.889
dow_cos	39.926
working_day	31.687
lag_7	24.373
lag_1	22.066
trend_days	19.048
lag_28	15.928
lag_14	15.236

feature	mean absolute shap
weekly_payday	3.836
monthly_payday	0.526
last_working_payday	0.479

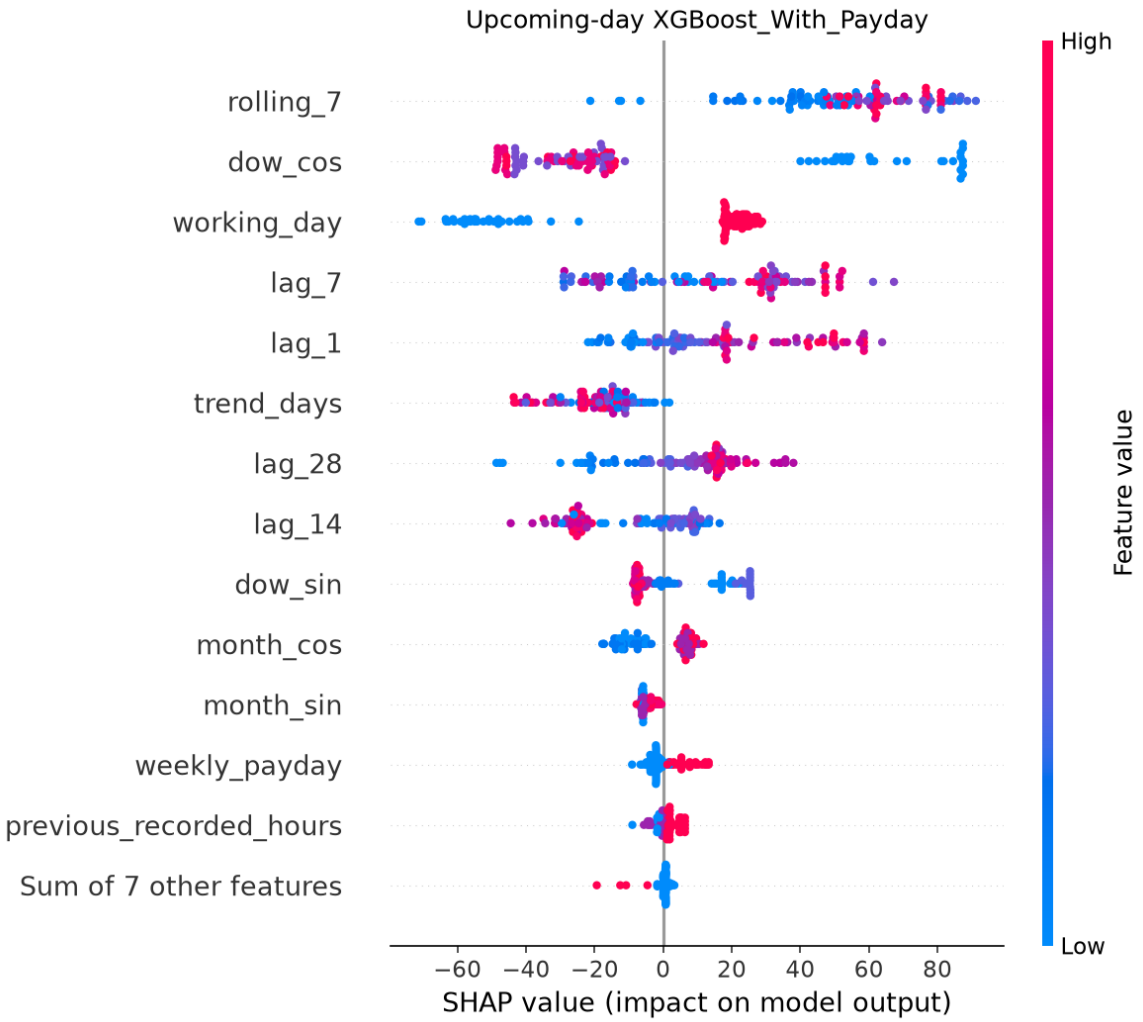


Figure 10. Daily XGBoost With_Payday Tree SHAP contribution distributions in GBP.

Tree SHAP additivity is numerically checked. Attributions explain fitted model predictions; correlated payroll/calendar flags share attribution. They do not identify actual customer incomes or prove causality. SHAP is applied to XGBoost, not asserted for every model.

Operational scenarios and limits

model	peak precision	peak recall	underforecast GBP	overforecast GBP
Prophet_Without_Payday	0.726	0.958	8866.274	10576.989
Prophet_With_Payday	0.729	0.972	8387.389	10949.765
XGBoost_Without_Payday	0.848	0.389	31528.144	2566.116
XGBoost_With_Payday	0.933	0.389	31968.387	2230.261
LSTM_Without_Payday	0.813	0.847	22759.171	5010.197
LSTM_With_Payday	0.831	0.819	22724.510	4907.509
Hybrid_Without_Payday	0.785	0.861	19188.979	5773.529
Hybrid_With_Payday	0.765	0.861	18981.275	5686.910
Sequential_Hybrid_Without_Payday	0.687	0.944	3585.269	25889.361
Sequential_Hybrid_With_Payday	0.690	0.958	3102.821	24774.100
Naive	0.803	0.792	14604.260	13421.600
Weekly_Seasonal_Naive	0.829	0.806	12773.040	8136.850

The training-only 75th-percentile daily revenue threshold defines a high-revenue workload scenario. Precision/recall evaluate forecasts flagging such days, and asymmetric under/overforecast revenue totals quantify planning risk. No shift scheduling, required staff counts, inventory reorder quantities, stockout savings or profitability improvement is claimed. Revenue is an imperfect workload proxy; actual operational evaluation requires transaction service times, staffing, SKU inventories, lead times and cost data. Department profiles support exploratory planning but cannot replace those inputs.

model	n	peak hour precision	peak hour recall
Prophet_Without_Payday	1622	0.600	0.008
Prophet_With_Payday	1622	0.545	0.008
XGBoost_Without_Payday	1622	0.620	0.578
XGBoost_With_Payday	1622	0.612	0.587
LSTM_Without_Payday	1622	0.595	0.229
LSTM_With_Payday	1622	0.594	0.172

The preserved one-hour forecasts also flag high-revenue hours using a separate training-only hourly 75th percentile. This directly explores peak-hour staffing signals, but revenue does not measure queue length or service time. Hourly and daily planning thresholds are distinct.

Research literature and methodological rationale

Taylor and Letham (2017 preprint; subsequently published 2018), Forecasting at Scale, presents a decomposable forecasting approach for analyst-configurable trend and seasonality. It motivates Prophet as an interpretable benchmark, but does not establish this store's payday window. https://peerj.com/preprints/3190.pdf

Chen and Guestrin (2016), XGBoost: A Scalable Tree Boosting System, describes regularised tree boosting and scalable computation. It motivates nonlinear tabular lag/calendar modelling; scalability does not imply generalisation on a single store. https://arxiv.org/abs/1603.02754

Hochreiter and Schmidhuber (1997), Long Short-Term Memory, develops gated recurrent memory for learning temporal dependencies. The present seven-day representation is a research choice, not a guarantee that an LSTM outperforms simpler methods with a small daily sample. https://doi.org/10.1162/neco.1997.9.8.1735

Lundberg and Lee (2017), A Unified Approach to Interpreting Model Predictions, frames additive feature contributions. It supports explanation of model associations, not causal payday inference. https://arxiv.org/abs/1705.07874

Hyndman and Athanasopoulos, Forecasting: Principles and Practice, describes time-series cross-validation with historical forecast origins and forecast accuracy measures. Chronological evaluation matches information availability; percentage errors are problematic at zero. https://otexts.com/fpp3/tscv.html and https://otexts.com/fpp2/accuracy.html. These methodological sources support design decisions; they do not substantiate sweeping slide claims that all previous retail work uses daily data or single models. A systematic payday/independent-retail literature review remains an academic task.

Holiday source: GOV.UK Bank Holidays API, https://www.api.gov.uk/gds/bank-holidays/ ; frozen JSON snapshot and SHA256 are saved for reproducibility. Sources consulted 8 October 2026.

Resolved work and remaining evidence requirements

Executed extensions provide direct day forecasts, a genuine hybrid ensemble, holiday/working-day predictors, configurable payroll scenarios, category behavioural outputs, positive-target MAPE, block-dependent uncertainty, expanding XGBoost validation, multi-seed LSTM evaluation, operational proxies and identifier-minimised outputs. No original raw records or research objectives were replaced.

A separate sequential residual cascade, sourced online employer-payroll experiment, targeted critical literature review and proposed governance/operational protocols are now implemented and documented in research_completion.md. Official employer schedules are not customers' actual paydays. Still requiring external evidence: customer income exposure, complete recording/opening hours, realised inventory/staffing/profit measures and institutional/store governance confirmation. Dissertation work is deferred at the researcher's request. No systematic-review completeness or deployment success is claimed.

Reproduction

From project_by_gpt use python run_extensions.py for additions only, or python run_all.py for both hourly and daily workflows. After daily preparation and base models, run src/hybrid_model.py, src/sequential_hybrid_model.py, src/online_payroll_analysis.py, src/privacy_export.py, src/next_day_comparison.py, src/extension_validation.py, src/research_completion.py and src/extension_report.py, followed by src/research_notebook.py. Comparison/reporting load saved forecasts and do not retrain. Cached holidays and frozen payroll source records support offline reruns.